

Presentation Script — Slides 8–18

Slide 8 — Decision Tree (Model Description)

Our second model was the **Decision Tree**, a **rule-based, nonlinear** model that is easy to interpret but tends to **overfit**.

The most important processing step here was `class_weight='balanced'`. Without it, the tree barely detected any Viral Reels; once applied, it finally started identifying the minority class. We then tuned the tree with **GridSearchCV**, controlling **depth** and **ccp_alpha pruning** to limit overfitting.

The key takeaway: **further tuning gave almost no additional gain**, and a **simpler, shallower tree actually generalized better** — an early sign that the limitation was not the model's complexity.

Slide 9 — Decision Tree (Results)

In the end, the Decision Tree reached a **higher Accuracy** than Logistic Regression, but its **ROC-AUC stayed around 0.5**. So despite being a more flexible model, it still **could not meaningfully separate Viral from Non-Viral Reels**.

Slide 10 — AdaBoost (Model Description)

Next was **AdaBoost**, a **sequential boosting** model that trains weak learners one after another, each focusing on the mistakes of the previous one.

The important detail: AdaBoost **does not support class_weight directly**, so we handled the imbalance using **balanced sample weights** instead, with a **shallow decision stump** as the base learner.

Slide 11 — AdaBoost (Results)

The impact of that sample weighting was dramatic — but in a misleading way. The model achieved an **extremely high Recall**, yet its **Accuracy collapsed**. What actually happened is that it began **predicting almost every Reel as Viral**, producing a flood of **false positives**. Its ROC-AUC again sat at **0.5**, so it learned **no real decision boundary**.

Slide 12 — Random Forest (Model Description)

Our fourth model was the **Random Forest**, an **ensemble of many Decision Trees** that reduces overfitting through **bagging and averaging**.

We used the **same preprocessing pipeline** with `class_weight='balanced'` and validated with **cross-validation**.

.

Slide 13 — Random Forest (Results)

Here we saw the **clearest symptom of the class imbalance**. The Random Forest reported a **high Accuracy**, but its **Recall and F1 were essentially zero** — meaning it achieved that accuracy simply by **labeling everything as Non-Viral**, the majority class. Combining many trees **did not improve separability at all**.

.

Slide 14 — XGBoost (Model Description)

Finally, our most advanced model: **XGBoost**, a **gradient boosting** model with built-in **regularization**, generally one of the strongest performers on **tabular data**.

For the imbalance, instead of a class weight we used `scale_pos_weight`, and added **early_stopping** to control overfitting. We then ran a **GridSearchCV** over depth, learning rate, and the number of trees.

.

Slide 15 — XGBoost (Results)

This was our most telling result. Even with a **powerful, carefully tuned boosting model**, the **ROC-AUC barely moved from 0.5** and there was **no meaningful improvement over the simpler models**. If even XGBoost couldn't find a boundary, that strongly suggested the problem was **the data, not the algorithm**.

.

Slide 16 — Model Comparison

Putting all five models side by side, the pattern is impossible to miss: **regardless of the algorithm — linear, tree-based, or boosting — they all landed at a ROC-AUC near 0.5**.

Each model failed in its own way — Logistic Regression found **no linear signal**, AdaBoost **over-predicted Viral**, and Random Forest and XGBoost **defaulted to the majority class**. But they all hit **the same ceiling: performance no better than random guessing**. (넘기지 말고 그대로 그 페이지)

.

Slide 17 & 18 — Additional Experiment: Threshold Adjustment (80 → 60)

This led us to test one hypothesis: maybe the real issue was the **severe class imbalance**. So we **lowered the Viral threshold from 80 to 60**, raising the Viral share from about **20% to 40%** for a more balanced dataset.

넘기기

The result was revealing. **F1-scores improved across several models**, which looked encouraging at first — but the **ROC-AUC stayed right around 0.5 for every model**. In other words, better balance produced **higher scores but no real ability to discriminate**.

This confirmed our core conclusion: **the limitation was never the model or the class balance — it was the dataset itself**. And that's exactly what we explore next in the Advanced EDA.

.

[Hand-off to next presenter / EDA section]